

A Multilingual Legal Chatbot Using Retrieval-Augmented Generation

Architecture Design and Pipeline Description

Abstract

Large language models are prone to generating plausible but factually incorrect information, a phenomenon known as hallucination. In the legal domain, such errors carry serious consequences. This paper presents the architecture of a multilingual legal chatbot built on a Retrieval-Augmented Generation (RAG) pipeline. The system accepts queries in Arabic, French, and English, retrieves relevant passages from indexed legal corpora using a hybrid dense-sparse retrieval strategy with Reciprocal Rank Fusion, and generates grounded answers via a large language model. Key design decisions include LLM-based zero-shot intent classification, structure-aware document chunking that preserves legal article boundaries, and a post-generation faithfulness verification module. The architecture addresses the specific challenges of multilingual legal information retrieval, including code-mixed queries, exact statutory reference matching, and implicit conversational references.

1. Introduction

Accessing legal and administrative regulations within academic institutions poses a significant challenge. Legal texts are written in complex language, distributed across heterogeneous documents, and often available in multiple languages. Users who need to understand specific procedures—such as doctoral enrollment requirements or administrative steps following exceptional circumstances—must manually search through numerous documents, a process that is both time-consuming and error-prone.

Conversational AI systems based on large language models (LLMs) offer a natural interface for such queries. However, LLMs are known to hallucinate: they may produce legally incorrect information with high apparent confidence. In a legal context, an erroneous answer can directly affect a user's rights and administrative procedures.

Retrieval-Augmented Generation (RAG) addresses this problem by grounding the generation process in externally retrieved documents rather than relying solely on the model’s parametric knowledge. This paper describes the complete architecture of a multilingual legal chatbot built on a RAG pipeline. The system supports Arabic, French, and English, and is designed to handle the specific challenges of legal information retrieval, including exact statutory references, multilingual code-mixing, and multi-turn conversational context.

The main contributions of this work are:

1. A hybrid retrieval strategy combining dense semantic search with sparse lexical matching via Reciprocal Rank Fusion (RRF).
2. An LLM-based zero-shot intent classification module that natively supports all three target languages without handcrafted rules.
3. A structure-aware chunking strategy that preserves legal article boundaries.
4. A post-generation faithfulness verification module that acts as a final safeguard against hallucinated legal content.

2. Related Work

Retrieval-Augmented Generation. Lewis et al. [1] introduced the RAG framework, which combines a parametric language model with a non-parametric retrieval component. The retriever fetches relevant passages from an external corpus, and the generator conditions its output on both the query and the retrieved context. This approach has become the standard architecture for knowledge-intensive NLP tasks where factual grounding is critical.

Hybrid Retrieval. Pure semantic retrieval using dense embeddings excels at capturing conceptual similarity but struggles with exact term matching. Conversely, sparse methods such as BM25 [2] handle exact lexical matches effectively but miss paraphrases and synonyms. Bruch et al. [4] demonstrated that combining both approaches yields consistent improvements. Reciprocal Rank Fusion (RRF) [3] provides a parameter-light method for merging ranked lists from heterogeneous retrieval systems.

Multilingual Embeddings. Sentence-BERT [5] established the use of siamese networks for producing sentence embeddings. More recently, BGE-M3 [6] introduced a model capable of generating both dense and sparse representations simultaneously across over 100 languages, making it particularly suitable for multilingual retrieval pipelines.

Evaluation of Generated Text. Traditional metrics such as BLEU and ROUGE rely on lexical overlap and perform poorly in multilingual settings. BERTScore [7] computes semantic similarity using contextual embeddings and has been shown to correlate better with human judgments across languages.

3. System Architecture

The system follows a sequential pipeline architecture consisting of ten modules. Table 1 summarizes the complete processing flow from user query to verified response.

#	Module	Description
1	Language Detection	Identifies query language (Arabic, French, or English)
2	Contextual Rewriting	Resolves implicit references using conversation history
3	Intent Classification	LLM zero-shot classification into three intent categories
4	Ambiguity Detection	Secondary LLM call when classification confidence is low
5	Query Routing	Directs the query to the appropriate retrieval collection
6	Hybrid Retrieval	Dense (BGE-M3) + sparse (BM25) search with RRF fusion
7	Deduplication	Jaccard-based near-duplicate removal
8	Reranking	Cosine similarity reranking, top-5 selection
9	LLM Generation	Context-grounded multilingual response generation
10	Faithfulness Check	LLM-based verification of generated claims against context

Table 1: Complete pipeline overview.

4. Methodology

This section describes each pipeline module in detail, including the design rationale and the techniques employed.

4.1 Language Detection

The system supports three languages: Arabic, French, and English. Detection follows a two-stage strategy.

Stage 1 — Unicode Script Analysis. Arabic script is detected by computing the ratio of Arabic Unicode characters (ranges U+0600–U+06FF, U+0750–U+077F, U+08A0–U+08FF, U+FB50–U+FDFF, U+FE70–U+FEFF) to the total number of alphabetic characters. If this ratio exceeds 0.30, the query is classified as Arabic. The threshold of 0.30 accounts for the frequent presence of French or English loanwords in modern Arabic text, consistent with thresholds used by established libraries such as `langdetect` and `pycld2` (typically 0.25–0.40).

Stage 2 — Statistical Detection. For non-Arabic text, the `langdetect` library is used. It employs character-level trigram language models and selects the language that maximizes the posterior probability given the input text.

4.2 Contextual Query Rewriting

In multi-turn conversations, follow-up queries often contain implicit references (e.g., “What about the deadlines?” following a discussion about doctoral enrollment). Before classification and retrieval, the system rewrites such queries by incorporating relevant context from the conversation history, producing a self-contained query suitable for single-turn retrieval.

4.3 Intent Classification

The system classifies each query into one of three intent categories:

- `legal_query`: questions about laws, regulations, or administrative procedures.
- `document_query`: questions about user-uploaded documents.
- `general_knowledge`: questions that do not require document retrieval.

Classification is performed via a zero-shot LLM call with a structured prompt that instructs the model to output exactly one intent label. This approach replaces a previous regex-based

heuristic system that suffered from three limitations: (1) regex patterns were written primarily in English and French, causing Arabic queries to be misclassified by default; (2) the scoring formula relied on empirically unvalidated constants; and (3) adding support for new phrasings required manual pattern engineering.

The LLM-based approach natively handles all three languages, requires no handcrafted rules, and introduces no arbitrary numerical parameters.

Design rationale: By delegating classification to the LLM, the system inherits the model’s multilingual competence. This eliminates the need for language-specific pattern maintenance and removes all manually tuned scoring constants from the classification pipeline.

4.4 Query Routing

Based on the classified intent, the query is routed to the appropriate data source:

- `legal_query` → Qdrant vector database, `legal_documents` collection.
- `document_query` → Qdrant vector database, `document_chunks` collection.
- `general_knowledge` → Direct LLM generation without retrieval.

4.5 Vector Representation

Document chunks and queries are encoded using BGE-M3 [6], a 2024 multilingual embedding model that produces both dense and sparse vector representations simultaneously.

Aspect	Previous Model	Current Model
Name	<code>mpnet-base-v2</code>	BAAI/BGE-M3
Year	2019	2024
Outputs	Dense 768d only	Dense + Sparse simultaneously
Arabic support	Limited	Advanced (extended corpus)
Reference	Reimers & Gurevych [5]	Chen et al. [6]

Table 2: Comparison of embedding models.

4.6 Hybrid Retrieval with Reciprocal Rank Fusion

A purely semantic retrieval approach fails on queries containing exact statutory references. For example, the embeddings for “Article 43” and “Article 44” occupy nearly identical regions in the vector space, making it impossible to distinguish them through cosine similarity alone. Lexical methods such as BM25 handle this case trivially through exact token matching.

Failure case: For a query such as “*Article 43 of Executive Decree 05-315*”, dense embeddings encode “Article 43” and “Article 44” in quasi-identical vector regions. Semantic search alone cannot distinguish these exact references. BM25 resolves this trivially via exact token matching.

The system therefore employs a true hybrid retrieval strategy with two components:

Dense retrieval uses the BGE-M3 dense embeddings to capture semantic similarity via cosine distance over the Qdrant index.

Sparse retrieval uses BM25 scoring with standard parameters ($k_1 = 1.5$, $b = 0.75$) as established by Robertson and Zaragoza [2].

The two ranked lists are merged using Reciprocal Rank Fusion [3]:

$$\text{RRF}(d) = \frac{1}{k + \text{rank}_{\text{dense}}(d)} + \frac{1}{k + \text{rank}_{\text{sparse}}(d)}$$

where $k = 60$ is the standard constant from the original RRF paper. This fusion method requires no tuning of interpolation weights and has been shown to outperform individual ranking methods as well as learned fusion approaches [3].

4.7 Deduplication

Hybrid retrieval may return near-duplicate passages retrieved by both the dense and sparse components. Duplicates are identified using the Jaccard similarity coefficient over token sets:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

Pairs with $J(A, B) \geq 0.85$ are considered duplicates, and the lower-scoring passage is removed. The threshold of 0.85 follows established practice in web-scale deduplication [8].

4.8 Reranking and Context Construction

The deduplicated candidate set is reranked by cosine similarity between the query embedding and each passage embedding:

$$\text{score}(q, d) = \frac{q \cdot d}{\|q\| \cdot \|d\| + \varepsilon}$$

where $\varepsilon = 10^{-9}$ ensures numerical stability. The top-5 passages are selected for context construction.

To ensure coverage across multiple source documents, a balanced selection policy guarantees a minimum of 2 chunks per source document before filling remaining slots by score.

4.9 LLM Generation

The selected passages, the rewritten query, and a summary of the conversation history are assembled into a structured prompt. The LLM is instructed to generate a response grounded in the provided context, cite specific legal articles where applicable, and respond in the same language as the user’s query.

4.10 Faithfulness Verification

As a final safeguard, the generated response is verified against the retrieved context through a secondary LLM call. The verification prompt presents both the context and the generated response, and asks whether every factual claim in the response is supported by the context. The model outputs a binary judgment: `faithful` or `not_faithful`.

If the response is judged unfaithful, the system returns a fallback message indicating that a reliable answer could not be found in the available legal documents. This mechanism ensures that the system fails safely rather than presenting ungrounded legal information with false confidence.

Safety principle: A legal chatbot that presents hallucinated information with confidence is more dangerous than one that acknowledges uncertainty. The faithfulness check constitutes the last line of defense against legal hallucinations.

5. Document Processing

5.1 Structure-Aware Chunking

Legal documents have a well-defined hierarchical structure (parts, chapters, articles, paragraphs). A fixed-size chunking strategy risks splitting an article across two chunks, destroying its semantic coherence. The system employs a hierarchical chunking strategy that respects document structure:

Priority	Boundary Type	Detection Patterns
1	Legal article boundaries	Article X , Art. X (French); Arabic article markers
2	Paragraph boundaries	Double line breaks
3	Sentence boundaries	Terminal punctuation (. ?)
4	Token-level fallback	512 tokens with 64-token overlap (12.5%)

Table 3: Hierarchical chunking priority levels.

The fallback chunk size of 512 tokens corresponds to the input limit of BERT-family models. The 12.5% overlap ensures contextual continuity between adjacent chunks, following the practice established in the original RAG work [1].

5.2 Chunking Parameters

Parameter	Value	Justification
Strategy	Structure-aware	Preserves legal article boundaries
Chunk size	512 tokens	BERT-family model input limit
Chunk overlap	64 tokens (12.5%)	Contextual continuity [1]
Top-K	5	LLM context budget, standard RAG practice
Min. per document	2 chunks	Multi-document coverage guarantee

Table 4: Document chunking and retrieval parameters.

6. Evaluation

The system is evaluated along two axes: retrieval quality and generation quality.

6.1 Retrieval Metrics

Three standard information retrieval metrics are used:

Precision@k measures the fraction of retrieved documents in the top- k results that are relevant. **Recall@k** measures the fraction of all relevant documents that appear in the top- k results. **Mean Reciprocal Rank (MRR)** captures how quickly the first relevant document appears in the ranked list, computed as the average of the reciprocal of the rank of the first relevant result across all queries.

6.2 Generation Metrics

BERTScore [7] is used as the primary automated generation metric. It computes token-level cosine similarities between contextual embeddings of the generated and reference texts, providing a semantic similarity measure that is more robust across languages than lexical overlap metrics such as BLEU or ROUGE.

6.3 Human Evaluation

Automated metrics are complemented by human evaluation along five dimensions:

Criterion	Description
Legal correctness	Whether the legal information provided is accurate
Completeness	Whether all relevant aspects of the question are covered
Clarity	Whether the response is understandable by a non-expert
Source citation	Whether references to specific legal texts are provided
Safe fallback	Whether the system appropriately declines when evidence is insufficient

Table 5: Human evaluation criteria.

7. Discussion

The architectural decisions described in this paper address several specific challenges in multilingual legal information retrieval.

Hybrid retrieval resolves the fundamental tension between semantic understanding and exact reference matching. Legal queries frequently contain precise statutory identifiers (e.g., “Article 43 of Executive Decree 05-315”) that dense embeddings cannot reliably distinguish. The combination of dense and sparse retrieval via RRF ensures that both query types are handled effectively without introducing tunable interpolation weights.

LLM-based intent classification eliminates the maintenance burden and language coverage gaps inherent in rule-based approaches. The previous regex-based system required manual pattern engineering for each language and relied on scoring constants that had not been validated on real user data. The zero-shot LLM approach handles all three languages natively and introduces no arbitrary parameters.

Structure-aware chunking preserves the semantic integrity of legal articles, which are the fundamental units of legal reasoning. Splitting an article across chunks risks losing critical context, such as conditions or exceptions stated later in the same article.

Faithfulness verification provides a critical safety mechanism for a legal application. A system that presents hallucinated legal information with confidence is more harmful than one that acknowledges uncertainty. The binary faithful/unfaithful check with a safe fallback ensures that users are not misled.

Limitations. The system relies on the quality and completeness of the indexed legal corpus. Documents not included in the index cannot be retrieved. The faithfulness verification module adds latency through an additional LLM call. Furthermore, the current evaluation framework has not yet been applied to a large-scale user study, which would be necessary to validate the system’s effectiveness in practice.

8. Conclusion

This paper presented the architecture of a multilingual legal chatbot based on Retrieval-Augmented Generation. The system processes queries in Arabic, French, and English through a ten-stage pipeline that includes language detection, contextual query rewriting, LLM-based intent classification, hybrid dense-sparse retrieval with Reciprocal Rank Fusion, structure-aware document chunking, and post-generation faithfulness verification.

The design prioritizes factual grounding and safe failure modes, which are essential properties for any system operating in the legal domain. Future work includes conducting a comprehensive user study, exploring cross-lingual retrieval where the query language differs from the document language, and investigating fine-tuned reranking models to further improve retrieval precision.

8. References

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